

A Declared-Block Longitudinal–Transverse Margin Compiler for the Literal V59 Scalar

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Abstract

We give an exact longitudinal–transverse margin compiler for the literal V59 source scalar after an exhaustive nonempty coordinate partition has been declared. The literal block weights are one, so the probes contract inside each physical output lane before projection. Projection onto the flat unit of each declared block yields an exact scalar decomposition; the transverse remainder is bounded first by its true block norms and then by a projected coherence envelope inherited from TPC-250. Any independently certified external scalar therefore obeys a computable lower margin, and strict longitudinal dominance implies nonvanishing. An exact rational 8-coordinate operator replay gives $C_{\text{long}} = 11/2$, $Q_{\text{trans}} = -1$, $C_x = 9/2$, and $R_{\text{trans}} = R_{\text{coh}} = 1$; with $(F, E) = (4, 1/2)$, the external bound and the lower margin 4 are attained. Equality is insufficient for nonvanishing. The partition and flat direction are modeling choices, and no actual V59 coherence asymptotic or arithmetic advance is claimed.

1 Source lock and scope

Let $H = \mathbb{C}^I$, where I is finite and nonempty, with inner product conjugate-linear in the first argument. Declare an exhaustive disjoint partition

$$I = \bigsqcup_d J_d, \quad J_d \neq \emptyset,$$

and let P_d be its coordinate projections. The same complete partition indexes source blocks b and output blocks c . From the literal TPC-247 source operator we import only [3]

$$\beta_b = P_b \beta, \quad A_{cb} = P_c A_x P_b, \quad v_{cb} = A_{cb} \beta_b, \quad w_c = P_c w.$$

The actual scalar has $\lambda_{cb} = 1$. Hence

$$g_c := \sum_b v_{cb} = P_c A_x \beta, \quad C_x = \langle w, A_x \beta \rangle = \sum_c \langle w_c, g_c \rangle. \quad (1)$$

The hard partition is a modeling choice. Once it is declared, define the block-flat unit

$$u_c = |J_c|^{-1/2} \mathbf{1}_{J_c}.$$

This unit is canonical only relative to J_c . It is not a V59-canonical direction, and it is not the TPC-219 longitudinal object [4]. These distinctions prevent the finite decomposition below from acquiring an unproved arithmetic interpretation.

2 The declared-block compiler

Set

$$a_c = \langle u_c, w_c \rangle, \quad b_c = \langle u_c, g_c \rangle, \quad w_c^\perp = w_c - a_c u_c,$$

and, probe by probe,

$$m_{cb} = \langle u_c, v_{cb} \rangle, \quad v_{cb}^\perp = v_{cb} - m_{cb} u_c, \quad g_c^\perp = \sum_b v_{cb}^\perp.$$

Let $G_c(bb') = \langle v_{cb}, v_{cb'} \rangle$. For $d_{cb} = \|v_{cb}^\perp\|$ define

$$D_c = \sum_b d_{cb}^2, \quad L_c = \sum_b d_{cb}.$$

If fewer than two d_{cb} are nonzero, set $\mu_c = 0$. Otherwise set

$$\mu_c = \max_{\substack{b \neq b' \\ d_{cb}d_{cb'} > 0}} \frac{|\langle v_{cb}^\perp, v_{cb'}^\perp \rangle|}{d_{cb}d_{cb'}}.$$

This is exactly the total empty-pair convention needed to apply TPC-250 to the projected family [1]. Finally put

$$U_c = \sqrt{D_c + \mu_c(L_c^2 - D_c)} \geq 0. \quad (2)$$

Theorem 1 (Literal declared-block margin compiler). *Define*

$$\begin{aligned} C_{\text{long}} &= \sum_c \overline{a_c} b_c, & Q_{\text{trans}} &= \sum_c \langle w_c^\perp, g_c^\perp \rangle, \\ R_{\text{trans}} &= \sum_c \|w_c^\perp\| \|g_c^\perp\|, & R_{\text{coh}} &= \sum_c \|w_c^\perp\| U_c. \end{aligned}$$

Then

$$C_x = C_{\text{long}} + Q_{\text{trans}}, \quad |C_x - C_{\text{long}}| \leq R_{\text{trans}} \leq R_{\text{coh}}. \quad (3)$$

Moreover, the projected Gram matrix is exactly

$$G_c^\perp(bb') = G_c(bb') - \overline{m_{cb}} m_{cb'}. \quad (4)$$

Proof. Exhaustiveness gives $\sum_b P_b = \sum_c P_c = I$, so (1) follows without tagged copies. Both $w_c^\perp$ and every $v_{cb}^\perp$ are orthogonal to u_c . Therefore

$$\langle w_c, g_c \rangle = \overline{a_c} b_c + \langle w_c^\perp, g_c^\perp \rangle,$$

where the conjugation of a_c is forced by the first-slot convention. Summation proves the identity in (3).

Expanding the two orthogonal projections and using $\langle v_{cb}, u_c \rangle = \overline{m_{cb}}$ gives (4). Applying the TPC-250 upper Gram bound to the projected probes with all weights one gives $\|g_c^\perp\|^2 \leq U_c^2$. Equation (2) chooses the nonnegative root, hence $\|g_c^\perp\| \leq U_c$. Cauchy–Schwarz followed by the triangle inequality now proves both inequalities in (3). $\square$

The theorem is a pointwise enclosure for fixed source data. It does not claim that varying an undeclared source parameter fills a disk, much less an exact disk image.

3 External margins and the strict endpoint

Corollary 2 (Conditional external compiler). *Let F be any independently certified scalar, and let $E \geq 0$ satisfy $|F - C_x| \leq E$. Then*

$$|F - C_{\text{long}}| \leq R_{\text{coh}} + E, \quad |F| \geq (|C_{\text{long}}| - R_{\text{coh}} - E)_+. \quad (5)$$

Consequently, $|C_{\text{long}}| > R_{\text{coh}} + E$ implies $F \neq 0$.

Proof. The first estimate follows from $F - C_{\text{long}} = (F - C_x) + (C_x - C_{\text{long}})$ and Theorem 1. Reverse triangle gives the second estimate after intersection with zero. A strict positive lower endpoint proves the final claim. $\square$

The hypothesis $|F - C_x| \leq E$ is an independent input. In particular, this paper does not obtain it automatically from the conditional synthesis interface of TPC-243 [2].

Proposition 3 (Equality obstruction). *In $\mathbb{R}^4$, let*

$$u = \frac{1}{2}(1, 1, 1, 1), \quad t = \frac{1}{2}(1, -1, 1, -1), \quad w = u + t, \quad g = u - t.$$

Then u, t are orthonormal and

$$C_{\text{long}} = 1, \quad Q_{\text{trans}} = -1, \quad R_{\text{trans}} = 1, \quad \langle w, g \rangle = 0.$$

Thus equality $|C_{\text{long}}| = R_{\text{trans}}$ cannot imply nonvanishing.

Proof. The two longitudinal coefficients equal one, while the transverse vectors are t and $-t$. Their covariance is -1 and their norm product is one, so the two scalar contributions cancel exactly. $\square$

4 Exact finite operator replay

The release certificate realizes a rational 8×8 operator. There are two blocks of size four; β is supported at the first coordinate of each block. In each block use

$$u = \frac{1}{2}(1, 1, 1, 1), \quad t_1 = \frac{1}{2}(1, -1, 1, -1), \quad t_2 = \frac{1}{2}(1, 1, -1, -1).$$

The two selected columns of A are assembled from

$$\begin{aligned} v_{00} &= \frac{1}{2}(1, 1, 1, 1) + \frac{3}{5}t_1, & v_{01} &= \frac{1}{4}(1, 1, 1, 1) + \frac{4}{5}t_2, \\ v_{10} &= \frac{1}{3}(1, 1, 1, 1) + t_1, & v_{11} &= \frac{2}{3}(1, 1, 1, 1) - t_1. \end{aligned}$$

Take

$$w_0 = \frac{1}{2}(1, 1, 1, 1) - \frac{3}{5}t_1 - \frac{4}{5}t_2, \quad w_1 = (1, 1, 1, 1).$$

c	D_c	L_c	μ_c	U_c	$\ g_c^\perp\ $	$\ w_c^\perp\ $
0	1	7/5	0	1	1	1
1	2	2	1	2	0	0

Table 1: Exact projected data for the synthetic finite operator replay. The second block has the largest coherence upper but contributes no transverse radius because both physical transverse vectors vanish after contraction or lane projection.

Direct rational calculation gives

$$C_{\text{long}} = \frac{11}{2}, \quad Q_{\text{trans}} = -1, \quad C_x = \frac{9}{2}, \quad R_{\text{trans}} = R_{\text{coh}} = 1.$$

For $F = 4$ and $E = 1/2$, both sides of the first estimate in (5) equal $3/2$, and the strict lower margin is 4. The replay is explicitly labeled *synthetic exact finite operator replay, not a literal V59 arithmetic instance*. It verifies the compiler and supplies no asymptotic evidence.

The real replay cannot detect a missing complex conjugation. The certificate therefore adds a Gaussian-rational two-coordinate fixture. It independently checks that $\bar{a}b = 1 - i$ rather than $ab = -1 + i$, and that one off-diagonal projected Gram entry is i only when $G_{12} - \bar{m}_1 m_2$ is used. Singleton and one-active-probe fixtures also verify $U = 0$ and the $\mu = 0$ empty-pair convention.

5 Computational trust boundary and limitations

The JSON producer uses exact rational arithmetic. The checker imports no producer code: it reparses canonical rationals, rejects duplicate keys, audits exact integer types and exhaustive blocks, recomputes the full operator replay, and rejects fifteen typed, semantic, stale-digest, duplicate-key, and digest-rebound mutations. A deterministic stress program checks 160 exact-rational declared partitions and probe families in normal and optimized Python modes. None of these finite checks proves an asymptotic statement.

The remaining mathematical gap is source-specific. The theorem neither controls the actual V59 projected coherences nor proves that the declared longitudinal term dominates every transverse and external loss. In particular, full Gate B and its global strict $1/400$ endpoint remain unpaid.

Claim firewall. The hard partition is an exhaustive-nonempty modeling choice, and the flat direction is relative only to its declared block. The TPC-243 external error is conditional, not automatic. Actual projected V59 coherence and payable longitudinal dominance are open. Arithmetic advance is no; fixed-atom credit is zero; L2 and the twin-prime result are none. Full Gate B is open, and its global strict $1/400$ endpoint is unpaid.

Maximum supported claim.

PROVED_STRUCTURAL_L1_LITERAL_V59_DECLARED_BLOCK_LONGITUDINAL_TRANSVERSE_MARGIN_COMPILER

6 Conclusion

The literal V59 source scalar admits an exact longitudinal–transverse split relative to every declared exhaustive nonempty coordinate partition. The projected Gram subtraction and TPC-250 coherence bound turn that split into a total, computable conditional margin, while the equality fixture identifies the necessary strict endpoint. The result is structural L1 progress only: the block-flat direction remains a modeling choice, the external error remains conditional, and the actual arithmetic margin remains open. There is no Route A claim, arithmetic advance, fixed-atom credit, L2 result, Gate-B closure, or twin-prime conclusion.

References

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